PRE-SEED · OCTOBER 2026

Oxiedo

Neural networks that report what broke while they train, and fix it.

oxiedo.com/black-box
Replay the archived run, component by component

Rokib Al Dhin Raadh, Founder
October 2026

Runs fail silently. No one can say which part broke.

Builders pay for failed runs in compute. Validators inherit a finished black box.

$170M

to train one frontier model (Llama 3.1, est.)

Every restart re-spends that compute.

∼20

loss spikes in one PaLM run

Fix: rewind, skip data, retry. The cause is never named.

35+

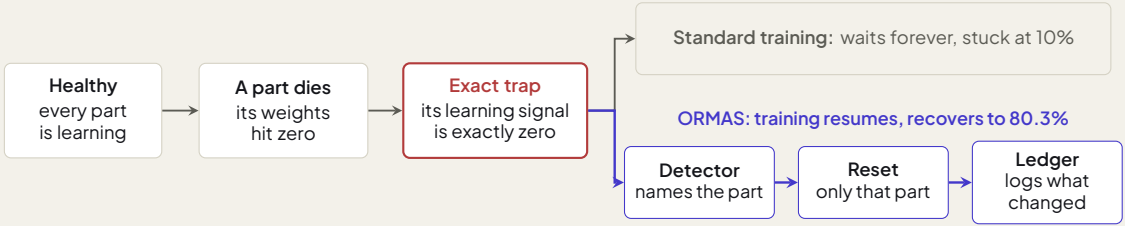
manual restarts of OPT-175B in 2 months

Divergence tracked a spike in **one layer**. Nothing named it as it happened.

$170M: Stanford AI Index 2025 (Llama 3.1–405B). 35+: Zhang et al., OPT-175B logbook, Meta 2022 (final-layer activations).
∼20: Chowdhery et al., *PaLM*, arXiv:2204.02311 (2022), §5.1 Training instability.

A dead part is a trap: we name, reset and record it.

Standard training cannot leave this trap. A targeted reset can, and every reset is written down.



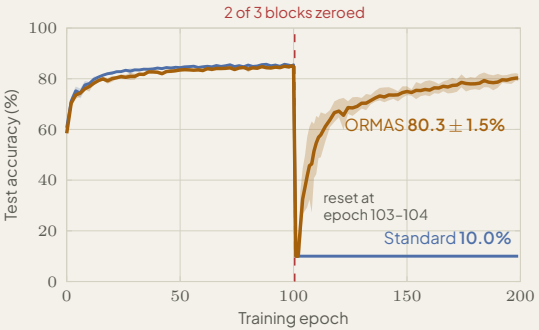

```
ledger | epoch 103 | block 1 | dead 3 epochs | reset
```

simplified from the seed-0 run's archived record

Trap: CIFAR-10 CNN, two of three blocks zeroed; standard runs stuck at chance in 12/12 archived runs (H-1, H-2). 80.3%: 3 seeds (H-1).
Ledger entry: seed-0 run, blocks flagged at epoch 101, reset in epoch 103 (F3 D-072).

Standard stays at 10%. ORMAS recovers to 80%.

Two of three blocks of a trained image model zeroed at epoch 100. Three seeds per curve.



WHAT HAPPENED

Standard: 10.0% on every seed. Its learning signal is exactly zero.

ORMAS names the dead blocks, resets them, recovers to 80.3%.

Every step is on the record: 85 logged interventions per run, each with block, epoch and reason.

Scope: CIFAR-10, one small CNN; curves are 3-seed means, band = seed min-max. Head-to-head vs other repair methods: A2.
Curves: archived runs `ablation_{standard,ormas}_cnn_dead_s0-2`, Jul 2026 (D-070/071).

Don’t trust us: replay the run yourself

oxiedo.com/black-box replays an archived run. Every figure is read from the archive.

THE RECORD · written during training

EPOCH	COMPONENT	DIAGNOSIS	ACTION	MAGNITUDE
101	conv block 1	dead	flagged	none
101	conv block 2	dead	flagged	none
103	conv block 1	dead	reset	
103	conv block 2	dead	reset	

Check it in two minutes

- Each component’s health, epoch by epoch.
- The ledger: component, epoch, reason, size.
- A scenario where ORMAS loses.

Narrated tour: Play the demo.

Try it now: open oxiedo.com/black-box, choose the dead-layer scenario, scrub to epoch 101 (the damage), then 103 (the reset).

Replay of archived run files; nothing is simulated in the browser. Rows: run ablation_ormas_cnn_dead_s0.

First product: a health monitor plus a change record

Runs beside their training job, read-only, on their machines. Already built; next is packaging.

TRAINING HEALTH MONITOR

Health of every component, on every run.

Names what failed, and when.

CHANGE RECORD

Every intervention: component, epoch, reason, size.

One file for the validator.

STATUS

Today: both run inside V1.

Next: record clean-up, then packaging.



Then, each gated by a named test: Label Audit (frozen-model test) · Source Separator (V2 separation study).

Record clean-up (V1.1): one format, exact steps, only applied changes logged, declared limits enforced. Commercial Plan 04, 06, 07.

Regulators now ask how an AI model was built

Four regulators now want evidence from training itself, not only from the finished model.

EU AI ACT · DEC 2027

Credit scoring is high-risk: document training, govern data.

Fines to €15M or 3% of turnover.

US · SR 26–2 · APR 2026

Replaces SR 11–7.

Non-generative AI models explicitly in scope, with ongoing monitoring.

UK · SS1/23

Deliberately covers AI models.

The Bank of England builds on it in 2026.

FDA · PCCP · DEC 2024

AI devices change only under a documented change plan.

1,614 AI devices listed.

Banks are not ready: large banks run 365 models; 67% use AI/ML, only 29% say their framework fits, and 75% put under 5% of staff on AI validation. Gartner: AI governance spend passes $1B by 2030.

EU: Reg. 2024/1689 Annex III 5(b), Art. 99; dates by Reg. (EU) 2026/1744. US: SR 26–2, 17 Apr 2026. UK: PRA SS1/23; BoE letter, 1 Apr 2026.
FDA: PCCP final guidance, 4 Dec 2024; AI device list, Sep 2026. Banks: Deloitte EMEA MRM survey 2025. Gartner, 17 Feb 2026.

A $19B market by 2035, and we know who signs

Model-risk spend is already a budget line. We sell into it from training.

$5.9B
AI model-risk management, 2025

$19.1B
by 2035; 12.5%/yr; 35% is finance

$149M
serviceable now, bottom-up

Who signs	Count	ACV	Annual	Next
ML platform leads: orgs training their own models	1,400+	$75K	$105M	M4: first 20 conversations
Heads of model risk: 66 US + 110 ECB banks	176	$250K	$44M	M4: 3 design partners

Adjacent: AI governance software $15.8B by 2030 (Forrester); MLOps $16.6B by 2030. **Expansion:** healthcare AI $110.6B by 2030; $145.5B top-20 pharma R&D. All inside **$2.7T** of AI spend in 2026.

SNS Insider, AI MRM, 25 Aug 2026; Gartner, 16 Sep 2026; Fed large banks, 30 Jun 2026; ECB list, Jul 2026; W&B >1,400 orgs.
Forrester 2024; Grand View 2025; MarketsandMarkets 2025; Deloitte 2024 (2023 spend). ACVs: our planned prices, untested. All: A8.

Trackers chart training; Oxiedo records each fix

Our axis: a per-component record of each fix, made in training, on their machines.

Buyer's alternative	Layer state in training	Names the part	Resets it	Record of each fix	Validator workflow
Rewind to checkpoint and retry	×	×	×	×	×
Trackers: W&B, MLflow	✓	~	×	×	~
Research resets: ReDo, Continual Backprop	✓	✓	✓	×	×
Data quality: cleanlab (finds bad labels)	×	×	×	×	×
MRM: ValidMind, ModelOp, Credo, Fiddler, Arthur	×	×	×	×	✓
Oxiedo (ORMAS)	✓	✓	✓	✓	~

The category is being bought: W&B → CoreWeave (~$1.7B, 2025) · Arize → Dynatrace ($915M, 2026) · Neptune → OpenAI (2025).

And funded: Arize $70M, Fiddler $30M, Credo AI $21M, ValidMind $8.1M (Point72). Detail: A10.

~ = partly. Vendor docs, 13 Oct 2026. ReDo: ICML 2023. CBP: *Nature* 2024. Deals: CoreWeave Mar 2025; Dynatrace 13 Aug 2026; OpenAI Dec 2025.

Ships fast, and tries to break the work first

Rokib Al Dhin Raadh, 18, Dhaka. Built V1 alone at 17.

DATED, CHECKABLE

Apr 2026	Own test suite killed the first design; rebuilt.
To date	383 controlled experiments, 4 model families.
Sep 2026	Replay demo shipped, oxiedo.com.
Sep 2026	DeepMath 2026 poster accepted.
Oct 2026	NeurIPS 2026 workshop reviewer.
Oct 2026	Ran every baseline reviewers asked for.
Earlier	AI-run e-commerce test: ~$6.7K in 12 months, ~95% AI-referred, $0 ads.*

GAP AND FILL

Solo. No regulated-buyer sales yet.

Month 1: research engineer. Month 7: a product engineer and a commercial lead.

Dhaka-based. The round funds Delaware incorporation; relocation follows the first design partner.

* Black Bloxie LTD (UK; father is director of record). One $4,386 order is ~65% of revenue. Tested an earlier agent system, not ORMAS.

Five milestones, each with a pass line written first

Research runs only where the product needs an answer. A failed milestone gets reported.

Milestone	Passes if	Month
M1 Repair benchmark	5 seeds vs. 3 published fixes and a dropout control; ours within 1 point of the best	2
M2 Detection study	Injected faults found at $\geq$90% precision and recall, flagged within 1 epoch	4
M3 Beyond CIFAR	M2's pass line holds on credit-risk and medical-image data	7
M4 Design partners	20 buyer conversations; 3 teams run the read-only monitor in-house	4 $\rightarrow$ 12
M5 Open record	Preprint and code public; NeurIPS 2027 submitted	5

Months from close, assumed Dec 2026; M5 is pinned to the NeurIPS 2027 deadline (mid-May). Tests: R4 §7; the reviewer asks they answer: A3.

$750K buys five tested milestones in 18 months

Dhaka cost base. Delaware C-corp at close; relocation follows the first design partner.

$750K
post-money SAFE

$7.5M
cap · 10.0% dilution

18 mo
Dhaka cost base

Use of funds	%	Milestones
Research engineer, remote, month 1	22	M1–M3, M5
Product and commercial hires, month 7	30	M4
Compute (rented GPUs)	10	M1–M4
Founder, living in Dhaka	6	all
Delaware C-corp, IP assignment, legal	9	close, M4
Partner travel, conferences	5	M4, M5
Tools and insurance	3	—
Buffer, unallocated	15	—
Total $750K	**100**	

Carta median cap for $250K–$1M SAFEs: ~$10M (2025). Programme terms (EF/Antler) replace the instrument. Detail: A5, A11.

The system of record for what training did

Every model should arrive with a record of what its training did, part by part.

Rokib Al Dhin Raadh, Founder
raadxbusiness9@gmail.com
oxiedo.com · raadh.me

Replay the run: oxiedo.com/black-box

Appendix: the questions that come next

Limits first, then the full record behind every number in the main deck.

A1	What is not established	Limits and the runs where ORMAS loses
A2	Full results	Every run, baselines, cost, the dropout control
A3	Falsification record	Kill rules, their outcomes, the venue record
A4	Pricing hypothesis	Licence, price band, renewal, gross margin
A5	Three-year model	Every row with its assumption; use of funds
A6	Entity and IP	Incorporation, IP assignment, code licence
A7	Sources	Every number, its source, its status
A8	Market evidence	Every market size, by analyst and date
A9	Buyer pain	Bank model risk, cost of failed runs, regulation
A10	Category deals	Acquisitions, raises, pricing comparables
A11	Round benchmarks	Carta caps, SAFE terms, accelerator deals

A1: What is not established, and where ORMAS loses

NOT ESTABLISHED

Scope. Every result is CIFAR-10/100 on small CNN, MLP and ResNet models. No transformer, tabular or customer data.

Repair. One seed: plain reset 83.3%, our controller 78.5%, ReDo 77.5%. What is sold is the detector and the record; M1 re-tests on 5 seeds.

Detection. Precision, recall and delay against known faults are unmeasured (M2).

V2 signal. The CNN disagreement pilot was negative: AUROC ≈ 0.51 vs plain loss ≈ 0.97.

Rebuild. The April suite that killed the predecessor has not been re-run on the current code.

Paper vs code. The paper states a 30% cap on resets; the code does not enforce it. Correction pending.

WHERE IT LOSES OR TIES

3 seeds, %	ORMAS	Plain
Random-direction injection	83.1	84.1
$\times 100$ weight explosion	85.1	86.0
ResNet-18 lesion	91.6	92.3
Clean CNN	85.6	85.1

Label noise 40%: ORMAS 75.6, CNN + dropout 74.5, plain 69.6. Most of the gain is dropout.

Cost: $\sim 2.6\times$ wall-clock (799 s vs 312 s); +15,867 parameters (+2.56%).

Register IDs: losses H-5; clean H-7; noise H-6; cost H-11/12; 1–seed baselines X-003; V2 pilot D-110/112; suite CP 21 risk 12; cap CP 21 V-1.

A2: ORMAS recovers what standard training cannot

Every archived lesion family, weak runs included.

Part zeroed at epoch 100	ORMAS	Comparison	Seeds	Reading
2 of 3 blocks, CIFAR-10 (headline)	**80.3 ± 1.5** (79.9 / 81.9 / 79.0)	Standard 10.0, every seed	3	Repair. Standard is frozen: every gradient is 0
Same lesion, Oct 4 rerun	78.5	**Plain reset 83.3**; ReDo 77.5; standard 10.0	1	Plain reset beats ORMAS by 4.8 pp (dropout confound)
3 of 3 blocks, CIFAR-10	71.5 ± 2.5 (72.9 / 73.1 / 68.6)	Standard 10.0, every seed	3	Repair, partial
1 block + 40% label noise	31.8 / 76.9 / 69.4	Standard 10.0, every seed	3	∼ Seed-variable; one seed reset the wrong block first
2 of 3 blocks, CIFAR-100	7.4 / 25.8 / 27.1 (pre-lesion 53.3)	Standard 1.0 (chance)	3	∼ Weak; still rising at epoch 199

Reading. Every win is a part whose learning signal is exactly zero; recovery starts at the reset (9 of 9 runs, epochs 103–104). ORMAS carries Dropout(0.3), the plain reset none: M1 reruns row 2 with ≥3 seeds and matched dropout.

Rows 1, 3–5: archived runs (H-1, H-5; F3 D-070, D-071, D-075). Row 2: RTX 3080, PyTorch 2.8, seed 0 (F0 X-003; D-131–134).
9 of 9: H-3. Wrong block first: D-065. All values: final test accuracy (%) at epoch 199.

A2: Clean, noisy and perturbed training

Clean data, label noise with the dropout control, perturbations and cost.

CIFAR-10 setting	ORMAS	Comparison	Seeds	Reading
Clean training	85.6 ± 0.4	Standard CNN 85.1 ± 0.5	3	$\sim$ Parity; epochs unequal
40% label noise, 200 epochs	75.6 ± 0.8	CNN + Dropout(0.3) 74.5 ± 0.4; plain 69.6 ± 0.9	5/3/3	$\sim$ Dropout explains $\sim$5.0 of 6.0 pp; rest +1.05, $p \approx 0.05$
40% label noise, 100 epochs	—	CNN + Dropout(0.5) 76.6 ± 0.8	3	Stronger dropout alone is as good or better
Random-direction weight injection	83.1 ± 0.3	Standard 84.1 ± 0.3	3	Loss, -0.9 pp; below standard every epoch
$\times 100$ weight explosion	85.1 ± 0.5	Standard 86.0 ± 0.1	3	Loss, -0.9 pp; batch norm absorbs the shock
ResNet-18, blocks behind a skip	91.6 ± 0.4	Standard 92.3 ± 0.3	3	Loss, -0.7 pp; the skip keeps signal flowing
Cost of the lesion run	799 s	Standard 312 s; reset 342 s; ReDo 452 s	1	$\sim 2.6\times$ wall-clock; +15,867 params (+2.56%)

Clean: H-7. Noise: H-6; F3 D-080/081 (Dropout 0.5: supp212). Injection ($5\times$ weight norm), explosion, ResNet: H-5, D-075.
Cost: RTX 3080, 1 seed (H-11, R-7); params on the 636,677-param CNN (H-12). Values: final test accuracy (%).

A3: We write the kill rule first, and follow it

Three pre-set tests went against us. Each one changed what we build.

When	Rule, written before the run	Outcome	What we did
Apr 2026	Predecessor architecture vs a purpose-built suite with a pass/fail list	Failed. Lost to a plain CNN at all four noise levels; zero corrections fired in 42 runs	Rewrote the system. The suite has not yet been re-run on the rebuild; that rerun is planned
Oct 3, 2026	V2 premise: disagreement between parts flags mislabelled samples (CNN pilot, 3 seeds)	Negative. AUROC ≈ 0.51, vs 0.97 for plain per-sample loss	V2 moves to parallel parts. The next pilot must beat loss and AUM by ≥ 0.03 AUROC, published either way
Oct 4, 2026	Paper rule: if ORMAS ends 2+ pp below a plain reset, do not submit	Triggered. ORMAS 78.5 vs reset 83.3 (-4.8 pp, 1 seed)	Paper held. V2 keeps only detect + reset; the product is the detector and the record

Why this is here. A single-author result is easy to tune until it flatters. Each rule above was written first, went against us, and stays on the record.

Apr 2026 suite: Commercial Plan 21 §2B (killtest_results, 2026-04-10). V2 pilot: F3 D-110/112. Paper rule: F0 X-003.
Next-pilot bar: F4 S-149. AUROC: how well a score separates mislabelled from clean samples (0.5 = chance).

A3: Every reviewer ask maps to a milestone

Venue record: accepted at DeepMath 2026; reviewer for NeurIPS 2026.

Venue	Outcome
DeepMath 2026	Accepted, poster (Sep 25); poster session Oct 29–30. Short abstract applying known stability results to training-time repair
NeurIPS 2026	Reviewer (Oct 3); Trustworthy AI for Good workshop, Paris (Dec 12)

What reviewers asked for	Status | milestone
Simple re-init and ReDo baselines	Done, 1 seed (A2); M1 adds seeds + dropout control
CBP and checkpoint-rollback baselines	M1
Leave-one-out ablations	Null on clean/noise (archived); under the lesion: M1
Compute cost	Done: ∼2.6× wall-clock (A2)
Detection precision, recall, delay	M2
Scale beyond CIFAR	M3
Stability assumptions, framing, jargon, dropout claim	M5 preprint, rewritten around A2

Venues: OpenReview decision, Sep 25, 2026 (DeepMath #39); NeurIPS 2026 reviewer recognition letter, Oct 3, 2026.
Reviewer asks: F2 Part F (R-1 to R-16), F4 S-156. Leave-one-out nulls: F2 H-8. Milestones: plan slide.

A4: An annual licence per team, on its own systems

THE LICENCE

Annual, per team, run on the customer's own machines or cloud.

No seats, no usage meter, no hosted tier: training data never leaves them.

Wedge: monitor and change record for an ML platform team.

Expansion: the same record, read by a model-risk validator.

PRICE BAND: HYPOTHESIS

Monitor: $75K/yr per ML team (band $75–250K).

Model-risk record: $250K/yr per validation team (band $200–500K).

Set against what a failed run already costs the team and what model-risk tooling costs the validator.

No price has been put to a buyer yet; M4 tests it.

WHY IT RENEWS

Records belong to the customer. They export every record, in a documented format, during the term and after it.

Renewal is earned on **new runs**: monitoring, updates, support, and a version their validator has reviewed.

Switching costs a re-validation, not lost history.

Gross margin logic. Nothing is hosted, so serving cost sits with the customer. Cost of revenue is deployment engineering and support: modelled at 70% in year 2 and 80% in year 3, to be replaced by pilot data.

Bands: Commercial Plan 14 (monitor) and 16 (model risk); same as s08. Plan figures, untested. Margins are A5 assumptions.

A5: Three-year model, year 1 at $0 revenue

$, year-end	Year 1	Year 2	Year 3	Assumption
Design partners (unpaid)	3	3	—	M4: 3 teams on the read-only monitor by month 12
Paying monitor teams	0	4	9	$75K/yr each; 1st converts by month 15
Paying model-risk teams	0	0	3	$250K/yr each; validator expansion from Y3
ARR	0	0.30M	1.43M	Teams × licence (A4 bands, as s08)
Revenue in year	**0**	0.15M	0.86M	Contracts land mid-year on average
Gross margin	—	70%	80%	No hosting; cost is deployment and support
Headcount	4	9	16	Founder + 3 hires from this round (A5 detail)
Operating spend	0.40M	1.40M	2.60M	Y1 = this round's plan; then ~$150K per loaded hire
Net burn	0.40M	1.30M	1.91M	Operating spend less gross profit
Money in	0.75M	3.0M	—	This round; seed at month 15 on M1–M5 (assumed)
Cash, year-end	0.35M	2.06M	0.14M	Series A in year 3; not modelled

No unit-economics ratio is shown. Without a paid contract, acquisition cost and lifetime value are unknown; the first three paid contracts set them.

Every figure is a planning assumption, not a forecast from data. Year 1 = months 1–12 after close. Licence band: A4.

A5: The $750K, line by line

Line	Mo 1–12	Mo 13–18	Total	%	Assumption → milestone
Research engineer, remote, month 1	110.0	55.0	165.0	22	$9.2K/mo loaded → M1–M3, M5
Commercial lead, month 7	60.0	60.0	120.0	16	$10K/mo → M4
Product engineer, remote, month 7	52.5	52.5	105.0	14	$8.75K/mo → M4
Compute, rented GPUs	50.0	25.0	75.0	10	Non-CIFAR data, pilots → M1–M4
Founder, living in Dhaka	30.0	15.0	45.0	6	$2.5K/mo → all
Delaware C-corp, IP, legal, accounts	35.0	10.0	45.0	6	Formed at close → M4 contracts
Regulatory, data-protection review	22.5	0.0	22.5	3	One engagement each → M4
Partner travel, conferences	25.0	12.5	37.5	5	Partner visits, NeurIPS → M4, M5
Tools, software, insurance	15.0	7.5	22.5	3	—
Buffer, unallocated	—	—	112.5	15	Slippage, or a 3-month bridge
Total, $K	**400.0**	**237.5**	**750.0**	**100**	

Seed timing. The seed is raised from month 12 on M1–M5 results (A5 model assumes month 15). The buffer covers three more months if it slips.

Current experiments cost about $200–600 of rented GPU time; the compute line budgets for non-CIFAR data, larger models and partner pilots.

A6: Oxiedo incorporates in Delaware at close

TODAY

Oxiedo is not yet incorporated. No cap table, no prior investor.

The founder personally holds the ORMAS code, run archive, paper and brand.

Black Bloxie LTD is a separate UK company; the founder's father is director of record. It ran the e-commerce experiment and is the interim contracting party for oxiedo.com.

AT CLOSE

Delaware C-corp formed before the SAFE is executed; cost is in the legal line (A5).

IP assignment at incorporation: code, archive, paper, brand and domain.

Any Oxiedo agreement or right held by Black Bloxie LTD moves to the C-corp; Black Bloxie takes no equity.

Founder stock vests. The founder, 18, is the contracting party.

CODE AND LICENCE

The code is not public today.

The architecture and run archive are **released on publication** under a research licence: free for research, teaching and evaluation. Production use is licensed (A4).

Not open source. An earlier MIT notice in the private repository is replaced before release.

Entity: README; oxiedo.com /about, /press. Licence: oxiedo.com /licensing. Bangladesh FX rules on US shares: counsel review before close.

A7: Every number and its source

Measured figures come from the run archive; market and price figures are labelled hypotheses.

Figure (slide)	Source	Status
10.0% vs 80.3 ± 1.5% (3, 4)	Archived lesion runs, 3 seeds each; F2 H-1, F3 D-070/071	Measured; rerun 4 Oct 2026
Stuck at chance 12/12; reset at epoch 103–104 in 9/9 (3, A2)	F2 H-2, H-3; archive-wide recount	Measured
Plain reset 83.3, ReDo 77.5, ORMAS 78.5 (4, A2)	Rerun, RTX 3080, PyTorch 2.8, seed 0; F0 X-003	Measured, 1 seed; M1 repeats it
383 experiments (10)	Run records across the founder's result folders (617 run IDs on disk)	Counted
∼20 loss spikes (2)	PaLM, arXiv:2204.02311 §5.1	Third-party paper
EU 2 Dec 2027; SR 26-2; SS1/23 (7)	Reg. (EU) 2026/1744; Fed SR 26-2; PRA SS1/23	Primary regulatory text
∼2.4 ×/yr training cost (7)	Cottier et al. (Epoch AI), arXiv:2405.21015	Third-party paper
$5.9B (2025) → $19.1B (2035) (8)	SNS Insider, AI model-risk management, 25 Aug 2026	Analyst report
1,400+ orgs; 176 banks; $149M (8)	W&B customer base; Fed large banks; ECB list; ACVs ours	Counts sourced; prices untested
$170M; 35+ restarts (2)	Stanford AI Index 2025; OPT-175B logbook (Meta, 2022)	Third-party
365 models; 29% (7)	Deloitte EMEA model-risk survey 2025	Third-party survey
$1.7B; $915M deals (9)	CoreWeave, Dynatrace announcements	Public filings/press
Competitor capabilities (9)	Vendor documentation, read 13 Oct 2026	Public claims
∼$6.7K, ∼95% AI-referred (10)	Black Bloxie LTD store records, 12 months	Founder records
$750K, $7.5M cap, use of funds (12, A5)	Carta pre-seed data 2025–26; our plan	Proposal

Registers: oxido/FINDINGS (F0 conversation, F2 paper claims, F3 data). Full results table: A2. Limits: A1.

A8: Analysts size the market in billions

AI model-risk management, its adjacent markets, and the spend around them.

Market	Size	Source, date
AI model-risk management	**$5.87B (2025) → $19.10B (2035)**, 12.5%/yr; finance 35%	SNS Insider, 25 Aug 2026
AI model-risk management	$5.7B (2024) → $10.5B (2029), 12.9%/yr	MarketsandMarkets, Jul 2024
AI model-risk management	$6.6B (2025) → $16.0B (2032), 13.5%/yr	P&S Intelligence, Aug 2026
AI model-risk management	$7.17B (2025), $8.33B (2026), $15B (2030)	The Business Research Co., Sep 2026
AI governance software	**$15.8B by 2030**, ~4× 2024	Forrester, 2024–25
AI governance platforms	$492M (2026) → >$1B (2030)	Gartner, 17 Feb 2026
AI trust, risk & security mgmt	$3.09B (2026) → $11.61B (2031), 30.3%/yr	MarketsandMarkets, Aug 2026
MLOps	$2.19B (2024) → $16.61B (2030), 40.5%/yr	Grand View Research, 2025
Data collection & labelling	$17.1B by 2030, 28.4%/yr	Grand View Research, Nov 2024
AI in healthcare	$21.66B (2025) → $110.61B (2030)	MarketsandMarkets, 2025
AI in military	$14.3B (2024) → $29.0B (2030)	Research and Markets, Jun 2025
Worldwide AI spend	**$2.7T (2026)**, +49.5%; infrastructure $1.43T	Gartner, 16 Sep / 19 May 2026
Big-4 hyperscaler capex	~$700–725B (2026), from ~$410B	Company guidance, Q1 2026
Top-20 pharma R&D	$145.5B (2023 spend)	Deloitte, 2024

Vendor market reports differ in scope; the deck's headline uses one series (SNS Insider) end to end. URLs: rework/research/R6.

A9: The buyer's pain is documented

Banks carry hundreds of models; large training runs fail as a matter of course.

Fact	Figure	Source, date
Models per bank (EMEA)	small 58 · medium 132 · **large 365**	Deloitte MRM survey, Oct 2025
Banks using AI/ML models	**67%** (90% of large); only 29% say their framework fits	Deloitte, Oct 2025
AI validation staffing	75% of banks put <5% of staff on AI validation	Deloitte, Oct 2025
Models per bank (US)	mean 175; cost (69%) and talent (56%) block more validation	RMA MRM survey, 2024
Cost to validate one framework	**$75,000 or more** (external)	RMA, Nov 2021
Validation time	1–17 weeks (US); up to 30 weeks (Europe)	McKinsey, ~2017
Frontier training cost	Llama 3.1–405B ~$170M; GPT-4 ~$79M; Gemini Ultra ~$192M	Stanford AI Index 2025
Cost growth	2.4× a year since 2016; $1B runs by 2027 on trend	Epoch AI, 2024
OPT-175B	35+ manual restarts, 100+ hosts cycled in 2 months	Meta logbook, 2022
Llama 3 405B	419 unexpected interruptions in a 54-day window (>90% effective)	Meta, arXiv:2407.21783
ByteDance MegaScale	job repaired and recovered >100 times; failures "the norm"	NSDI 2024
EU AI Act fines	€35M/7% prohibited; **€15M/3% high-risk**	Reg. 2024/1689 Art. 99
Risk-governance penalties	Citi: $400M (2020), $136M (2024), data and risk governance	OCC, Fed

Citi penalties concern risk and data governance, not model risk specifically. URLs: rework/research/R7.

A10: Buyers are paying for this category now

Training and model observability is being acquired and funded, and priced in five figures.

Deal or price	Value	Source, date
CoreWeave acquires Weights & Biases	∼**$1.7B** (reported); closed May 2025	CoreWeave; TechCrunch
Dynatrace to acquire Arize	**$915M**	Dynatrace, 13 Aug 2026
OpenAI acquires Neptune (training-run tracking)	undisclosed; reported <$400M in stock	OpenAI, Dec 2025
Cisco acquires Robust Intelligence; Galileo	undisclosed	Cisco, Sep 2024; Apr 2026
Snowflake acquires TruEra	undisclosed	Snowflake, May 2024
Arize Series C	$70M	TechCrunch, Feb 2025
Fiddler Series C	$30M (∼$100M total), regulated industries	Jan 2026
Credo AI	$21M ($41.3M total)	SiliconANGLE, Jul 2024
ValidMind seed, bank model risk	$8.1M, led by Point72 Ventures	FinTech Global, Mar 2024
W&B contract, median	**$42,900/yr** (52 purchases; $6.3K–$136K)	Vendr, 2026
ValidMind	∼$5,000 per model per year	AWS listing (secondary)
Credo AI	annual, per AI use case	AWS Marketplace
External validation, one framework	$75,000 or more	RMA, 2021

Our planned ACVs ($75K monitor, $250K model risk) sit between tool pricing and validation spend: A4. URLs: rework/research/R7.

A11: The round is priced inside market norms

$750K at a $7.5M cap sits below Carta's median for its size.

Benchmark	Value	Source, date
Median cap, $250K–$1M SAFEs	∼**$10M**	Carta, State of Pre-Seed 2025
Median cap, $1M–$2.5M SAFEs	∼$15M	Carta, State of Pre-Seed 2025
Median cap, SAFEs >$2.5M	$35M, +40% year on year	Carta, Q2 2026
Pre-seed rounds on SAFEs	93%	Carta, Q2 2026
Post-money SAFEs with a cap	94%	Carta, H1 2026
Cap-only SAFEs (no discount)	61% (8,762 SAFEs, 2024)	Carta
Median US pre-seed deal	∼$0.9M	PitchBook–NVCA via SaaStr, Q2 2025
SAFE volume, 2025	$10.4B across 50,316 instruments	Carta, Feb 2026
AI valuation premium	1.2–1.5× pre-money, seed to Series C	Gunderson Dettmer, 2025
Solo founders	36.3% of new startups (H1 2025)	Carta Solo Founders Report
YC	$500K: $125K for 7% + $375K uncapped MFN	Y Combinator
Antler Singapore	$150K: $100K for 10% + $50K MFN	Antler, 2026
Entrepreneur First	$250K: $125K for 8% + $125K MFN	third-party, 2026

Programme terms replace s12's instrument; milestones are unchanged. URLs: rework/research/R8.